DATABASE MANAGEMENT SYSTEMS LAB

Lab Report Submitted to

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LAB EXPERIMENT- 4

Objective

To study and implement user creation, deletion, and privilege management using SQL commands such as **CREATE USER, GRANT, REVOKE, and DROP USER**.

Database Creation

CREATE DATABASE

yukti_college_db;

Explanation

Creates a new database to store tables and data.

SE yukti_college_db;

Explanation

Selects the database so that all operations are performed inside yukti_college_db.

Table Creation

```
CREATE TABLE Student (
  student_id INT PRIMARY KEY,
  student_name VARCHAR(50),
  marks INT
);
```

Explanation

- Creates Student table
- student_id → unique identifier (**PRIMARY KEY**)
- VARCHAR(50) → stores text
- INT → stores numbers

```
CREATE TABLE Course (
  course_id INT PRIMARY KEY,
  course_name VARCHAR(50)
```

);

Explanation

- Creates **Course table**
- course_id uniquely identifies each course

Insert Values

INSERT INTO Student VALUES

(1, 'Yukti', 85),

(2, 'Neha', 90),

(3, 'Sonal', 78);

Explanation

- Inserts multiple records into Student table
- Each row represents a student

INSERT INTO Course VALUES

(101, 'DBMS'),

(102, 'OS'),

(103, 'Networks');

Explanation

- Adds course records into Course table

Part A: User Creation

(1) Create user1

CREATE USER 'user1'@'localhost' IDENTIFIED BY 'pass123';

Explanation

- Creates a new user **user1**
- 'localhost' means user can access from same system
- IDENTIFIED BY sets password

(2) Create user2

CREATE USER 'user2'@'localhost' IDENTIFIED BY 'pass456';

Explanation

- Creates another user with different credentials

Part B: Granting Privileges

(3) Grant SELECT and INSERT to user1

```
GRANT SELECT, INSERT
ON Student
TO 'user1'@'localhost';
```

Explanation

- Gives permission to:
  - SELECT → view data
  - INSERT → add data
- Applies only on **Student table**

(4) Grant ALL privileges to user2

```
GRANT ALL PRIVILEGES
ON Course
TO 'user2'@'localhost';
```

Explanation

- Gives full access (SELECT, INSERT, UPDATE, DELETE, etc.)
- Applies on **Course table**

(5) WITH GRANT OPTION to user1

```
GRANT SELECT, INSERT
ON Student
TO 'user1'@'localhost'
WITH GRANT OPTION;
```

Explanation

- **Gives same privileges as above**
- **WITH GRANT OPTION allows user1 to grant these permissions to other users**

C: Revoking Privileges

(6) Revoke INSERT from user1

```
REVOKE INSERT
ON Student
FROM 'user1'@'localhost';
```

Explanation

- Removes only **INSERT** permission
- User1 can still use SELECT

(7) Revoke ALL from user2

```
REVOKE ALL PRIVILEGES
ON Course
FROM 'user2'@'localhost';
```

Explanation

- Removes all permissions from user2 on Course table

Part D: User Deletion

(8) Drop user2

```
DROP USER 'user2'@'localhost';
```

8	01:59:02	CREATE USER 'user1'@'localhost' IDENTIFIED BY 'pass123'	0 row(s) affected	0.015 sec
9	01:59:26	CREATE USER 'user2'@'localhost' IDENTIFIED BY 'pass456'	0 row(s) affected	0.016 sec
10	01:59:34	GRANT SELECT, INSERT ON Student TO 'user1'@'localhost'	0 row(s) affected	0.000 sec
11	01:59:44	GRANT ALL PRIVILEGES ON Course TO 'user2'@'localhost'	0 row(s) affected	0.031 sec
12	01:59:51	GRANT SELECT, INSERT ON Student TO 'user1'@'localhost' WITH GRANT OPTION	0 row(s) affected	0.000 sec
13	01:59:59	REVOKE INSERT ON Student FROM 'user1'@'localhost'	0 row(s) affected	0.015 sec
14	02:00:09	REVOKE ALL PRIVILEGES ON Course FROM 'user2'@'localhost'	0 row(s) affected	0.000 sec
15	02:00:23	DROP USER 'user2'@'localhost'	0 row(s) affected	0.031 sec

Explanation

- Permanently deletes user2 from database
- User cannot login anymore

(9) Observation

After deleting user2, login attempts using its credentials will fail, showing an authentication error.

Part E

Additional Commands

(10) Display all users

```
SELECT user, host FROM mysql.user;
```

	user	host
▶	mysql.infoschema	localhost
	mysql.session	localhost
	mysql.sys	localhost
	root	localhost
	user1	localhost

Explanation

- Displays all users in MySQL system
- mysql.user is a system table storing user info

(11) Show privileges of a user

SHOW GRANTS FOR 'user1'@'localhost';

	Grants for user1@localhost
▶	GRANT USAGE ON *.* TO `user1`@`localhost`
	GRANT SELECT ON `yukti_college_db`.`studen...

Explanation

- Shows all privileges assigned to user1
- Useful for checking permissions

Result

The database and tables were created successfully. Users were created and assigned privileges using GRANT. Permissions were modified using REVOKE, and a user was deleted using DROP USER. This experiment demonstrated effective implementation of database security and access control mechanisms.